

Data-Driven Model Free Adaptive Perimeter Control for Multi-Region Urban Traffic Networks With Route Choice

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Abstract—Recent studies have shown that a homogenous urban road network exists a well-defined macroscopic fundamental diagram (MFD), which can be used for perimeter control conveniently. Most of the existing perimeter control strategies are model-based control methods, leading to the result that the control effect may be not good enough if the traffic model is not accurate. In this paper, a novel data-driven strategy called model free adaptive control (MFAC) method is proposed for multi-region perimeter control in order to get rid of the dependency of the aforementioned model-based MFD-based perimeter control methods. Due to the fact that the multi-region urban traffic system (MRUTS) is a complex interconnected system, a decentralized estimation and decentralized MFAC (DED-MFAC) method is utilized to deal with the strong-coupled characteristic of the traffic system. In this framework, MFD is used to determine the desired accumulations in each region and generate the throughput data of the urban traffic system, since the acquisition of trip completion flow is more difficult than accumulations. In addition, route choice is also integrated in the proposed policy to further improve the performance of the urban traffic system. A key advantage of the proposed approach is that it can only use the traffic data instead of the traffic model for real-time perimeter control. The effectiveness of the proposed perimeter control scheme is tested in simulation for a multi-region system, and the results show that it is superior to some other commonly used perimeter control methods.

Index Terms—Macroscopic fundamental diagram (MFD), decentralized estimation and decentralized model free adaptive control (DED-MFAC), route choice, real-time perimeter control.

I. INTRODUCTION

URBAN traffic system is indispensable to people's life. With the expansion of cities and the improvement of people's living standard, the vehicles travelling on the urban road increase dramatically. Meanwhile, the contradiction between the growing number of vehicles and limited infrastructure

becomes progressively prominent. Since the expansion of urban roads and other infrastructures is costly and the available space is limited, traffic control has attracted great interest of researchers and engineers.

Up to present, there are already a number of urban traffic control strategies, such as SCOOT [1], SCATS [1], [2], OPAC [3], TUC [4], [5], MPC [6], [7], etc. Most of the existing traffic control methods are based on minimizing the traffic delay, which is not easy to be obtained in practice. Meanwhile, they are difficult to handle the problem of the imbalanced traffic load, which often appears in reality. In recent years, some perimeter control methods are presented to balance the urban traffic load using macroscopic fundamental diagram (MFD).

MFD describes a unimodal, low-scatter relationship between the accumulation and the trip completion flow in homogeneous urban traffic networks. The original idea of MFD was provided in [8], and similar approaches were also introduced in [9], [10]. The existence of MFD was verified in the downtown of Yokohama, Japan [11], while a derivation of MFD via traffic flow is presented in [12]. These works indicated that MFD has the following three properties: (i) some homogeneous urban regions with suitable size approximately exhibit an MFD relating accumulation to space-mean flow, (ii) there is a robust linear relation between the region's space-mean flow and its trip completion flow (the rate that vehicles reaching their destinations, including leaving the region and finishing their trips in it), and (iii) although MFD is affected by the infrastructure and control strategies of the regions (see [13], [14]), it is not sensitive to the change of traffic demands. Property (i) is important for modeling purpose as detailed dynamics of individual links and intersections are no longer required. Property (ii) is helpful for monitoring purpose because the average flow can be measured more easily than the trip completion flow. Property (iii) is applicable for control purpose since the traffic administration can formulate traffic control and management measures without detailed information of O-D table.

Based on MFD, the perimeter control methods have been designed where the MFD is used to provide the desired volume of traffic accumulation. Perimeter control for single region can be found in [15]–[19]. Stability analysis of perimeter control for two-region networks is treated in [20], [21], and perimeter control for two urban regions is presented in [20], [22], [23]. For a multi-region urban traffic system (MRUTS), a perimeter feedback regulator with online adaptive optimization is

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presented in [24], a Linear-Quadratic-Integral (LQI) based feedback perimeter control strategy is introduced in [25], and a transfer function based perimeter control scheme can be found in [26]. Some other perimeter control methods for multi-region systems are shown in [27]–[29]. In order to improve the performance of perimeter control for the large-scale urban road network systems, the route guidance is also introduced into perimeter control [13], [30], and some other route guidance strategies can be found in [31], [32].

The above perimeter control methods are all based on the MFD-based traffic model, in which the homogeneity of each region is required. However, if the urban road network or the region is heterogeneous, the MFD will be not well-defined, especially under congested condition. It will have high scatter and hysteresis phenomena during the process of formation and dissipation of congestion [33]–[36], and further the inaccurate MFD-based traffic model will lead to the deterioration of the control effects of the above model-based perimeter control methods. In order to deal with this issue, some partitioning methods are presented in [37]–[41] to homogenize the heterogeneous urban road networks. Nonetheless, homogeneous regions are difficult to be partitioned perfectly, and the modeling process of MFD is also arduous, which makes the model-based perimeter control methods hard to be used. Another factor that could lead to the inaccuracy of traffic model is that vehicles always not fully comply with the route guidance. Furthermore, all the traffic information is required to be known to implement the above existing perimeter control methods, such as time-varying traffic demands, route choice ratios, and regional O-D tables, etc., which are difficult to be obtained. Finally, most of the above perimeter control strategies are centralized control methods, and the internal coupling of multiple-input and multiple-output (MIMO) nonlinear systems needs to be mathematically described accurately, which is difficult to deal with.

Nowadays, the urban transportation system produces numerous traffic data every day, and it will be of great significance to make use of traffic data instead of traffic model to deal with urban traffic problems, which can be realized by taking advantage of computer and network technology. Thus, the data-driven approaches has been promoted. The data-driven methods have been applied in many aspects of urban traffic field in the last decade, such as traffic flow prediction [42]–[44] and traffic control [45], [46], etc. For perimeter control, using traffic data for controller designing can avoid all the drawbacks of the model-based methods, such as modeling difficulty, robustness less, etc., so that the traffic control performance can be improved.

In recent years, model free adaptive control (MFAC) has been investigated extensively in the data-driven control community. MFAC is originally proposed in [47], and the systematic framework has been shaped and the stability of the closed-loop system has been proved in [48], [49]. MFAC scheme for MIMO systems is also presented and it has been successfully applied in many practical plants, such as freeway ramp metering [45], AC/DC microgrids [50], networked nonlinear systems [51], computer communication networks [52], and so on. A key advantage of MFAC is that the model

information of the controlled system is no longer needed. Instead, it merely needs the input and output data of the system to design the controller. As the MRUTS is essentially a MIMO nonlinear system, inspired by the above discussion, in this paper the decentralized estimation and decentralized MFAC (DED-MFAC) method [49] is utilized for perimeter control to deal with the coupling of the MRUTS.

The main contributions of this paper are as follows.

1) A novel description of coupled MIMO system and a system decomposition scheme for MRUTS are proposed for macroscopic traffic flow perimeter control of the heterogeneous urban road networks.

2) A novel data driven control method called DED-MFAC is first applied to perimeter control for large-scale multi-region complex connected urban traffic system. It only use the input and output data of the urban road network system to design the perimeter controller. Different from the existing centralized MFAC for MIMO systems, it is implemented with the decentralized estimation and decentralized control manner.

The advantages of this method are as below:

1) Comparing with the existing MFD-based perimeter control methods, the time-consuming and laborious work of MFD modeling, the precise partitioning of the urban traffic network, and the regional O-D tables, are all avoided in the proposed DED-MFAC perimeter control scheme. Instead, a sketchy partitioning according to geometrical location and other factors for the urban traffic network is enough, since the precise model is not necessity.

2) Comparing with the model-based MIMO perimeter control strategies, the difficulty of urban traffic modeling, the accurate description and treatment for the coupling of complex nonlinear systems for the MRUTS are also avoided. Instead, in this work, the coupling interactions among the different regions are only required to be measurable. In the MRUTS, the physical meaning of the coupling is that the transfer flows among the adjacent regions, which is easy to be measured with the advanced sensors, such as magnetic induction coil, traffic video detector, etc.

3) Additionally, route choice is combined with the proposed perimeter control strategy in order to make it more in line with the actual traffic situation and further enhance the efficiency of the urban traffic system, while the precise route choice ratios are not needed in the process of perimeter controller design.

The rest of this paper is organized as follows. The dynamics for MRUTS is formulated in Section II. In Section III, the DED-MFAC strategy for perimeter control of the MRUTS is described. Section IV depicts the numerical simulation results of the proposed DED-MFAC method compared with some other commonly used perimeter control strategies. The main conclusions and future works are summarized in Section V.

II. TRAFFIC DYNAMICS FOR MULTI-REGION URBAN TRAFFIC NETWORK SYSTEM

Assume that there is an urban traffic network $\mathcal{N}$ partitioned into N regions, i.e., $\mathcal{N} = \{1, 2, \dots, N\}$, which is diagrammed in Fig. 1, and that each region has an MFD with trip

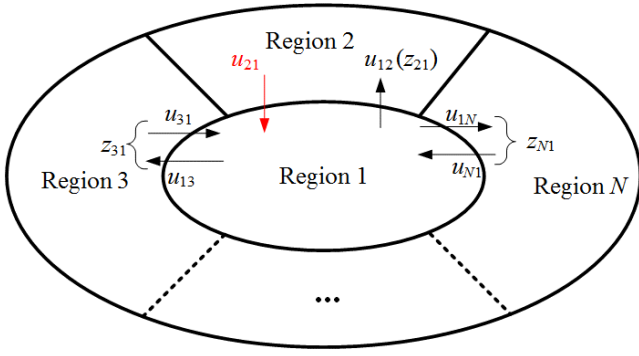


Fig. 1. An urban road network partitioned into N regions.

completion flow $G_i(n_i(k))$. In this section, the traffic dynamics for the MRUTS is given here.

A. Dynamics of Multi-Region Urban Traffic Networks

In order to save space, the first order Eulerian discretization of the urban traffic dynamics is directly given in the following. The traffic dynamics is similar to [13], [30]. It is noteworthy that the traffic model here is only utilized to generate traffic data instead of perimeter controller design. The dynamics of the multi-region urban traffic network is presented as follows:

$$n_{ij}^f(k+1) = \begin{cases} n_{ij}^f(k) + T \cdot (q_{ij}(k) + \sum_{\substack{g \in N_i, g \notin N_j \\ i \in V_{gj}}} u_{gi}(k) \cdot M_{gij}(k) - u_{ij}(k) \cdot M_{ij}^f(k)), & j \in N_i \\ n_{ij}^f(k) + T \cdot (q_{ij}(k) + \sum_{\substack{g \in N_i, g \notin N_j \\ i \in V_{gj}}} u_{gi}(k) \cdot M_{gij}(k) - \sum_{h \in V_{ij}} u_{ih}(k) \cdot M_{ihj}(k)), & j \notin N_i \end{cases} \quad (1)$$

$$n_{ij}(k) = \begin{cases} n_{ij}^f(k) + \sum_{\substack{h \in N \setminus N_i \\ j \in V_{ih}}} n_{ijh}(k), & j \in N_i \\ \sum_{h \in V_{ij}} n_{ihj}(k), & j \notin N_i \end{cases} \quad (2)$$

$$M_{ij}(k) = \frac{n_{ij}(k)}{n_i(k)} \cdot G_i(n_i(k)), j \in N_i \quad (3)$$

$$M_{ij}^f(k) = \frac{n_{ij}^f(k)}{n_i(k)} \cdot G_i(n_i(k)), j \in N_i \quad (4)$$

$$M_{ijh}(k) = \frac{n_{ijh}(k)}{n_i(k)} \cdot G_i(n_i(k)), j \in N_i, h \notin N_i, j \in V_{ih} \quad (5)$$

$$M_{ii}(k) = \frac{n_{ii}(k)}{n_i(k)} \cdot G_i(n_i(k)) \quad (6)$$

$$M_{ij}(k) = M_{ij}^f(k) + \sum_{\substack{h \notin N_i \\ j \in V_{ih}}} M_{ijh}(k), j \in N_i \quad (7)$$

$$n_{ij}(k) = n_{ij}^f(k), j \notin N_i \quad (8)$$

$$n_{ihj}(k) = b_{ihj}(k) \cdot n_{ij}(k), j \notin N_i, h \in V_{ij} \quad (9)$$

$$n_{ii}(k) = n_{ii}^f(k) \quad (10)$$

$$n_{ii}(k+1) = n_{ii}(k) + T \cdot (q_{ii}(k) + \sum_{j \neq i, j \in N_i} u_{ji}(k) \cdot M_{ji}^f(k) - M_{ii}(k)) \quad (11)$$

$$n_i(k) = n_{ii}(k) + \sum_{j \neq i} n_{ij}^f(k) \quad (12)$$

where the variables are defined as follows:

i, j, h, g	Symbols representing regions and subsystems,
N_i	The set of regions neighboring region i ,
V_{ij}	The set of regions that vehicles from region i to region j can go through next immediately,
$n_i(k)$	The total number of vehicles travelling in region i at time step k ,
$n_{ij}(k)$	Number of vehicles in region i travelling to region j at time step k ,
$n_{ij}^f(k)$	Number of vehicles in region i with final destination to region j at time step k ,
$n_{ii}(k)$	Number of vehicles in region i with destination to itself at time step k ,
$n_{ihj}(k)$	Number of vehicles in region i travelling to the next immediate region h with destination to region j at time step k ,
$q_{ij}(k)$	Exogenous traffic demands generated in region i with destination to region j at time step k ,
$q_{ii}(k)$	Endogenous traffic demands in region i at time step k ,
$M_{ij}(k)$	Transfer flow from region i to region j at time step k ,
$M_{ij}^f(k)$	Transfer flow from region i to region j with final destination to region j at time step k ,
$M_{ihj}(k)$	Transfer flow from region i to region h with final destination to region j at time step k ,
$M_{ii}(k)$	The internal flow from region i with destination in itself at time step k ,
$G_i(n_i(k))$	Trip completion flow for region i at time step k ,
$u_{ij}(k)$	Perimeter control ratio from region i to region j at time step k ,
$b_{ihj}(k)$	The ratio of vehicles in region i choosing to go through the next immediate region h to reach their destination region j at time step k ,
T	The sampling and the control cycle length.

Remark 1: It is noteworthy that the transfer flow $M_{ij}(k)$ and $M_{ij}^f(k)$ are only for $j \in N_i$. $M_{ij}(k)$ and $M_{ij}^f(k)$ are both equal to 0 for $j \notin N_i$. Under the effect of perimeter control,

$u_{ij}(k) \cdot M_{ij}(k)$, $u_{ij}(k) \cdot M_{ij}^f(k)$, and $u_{ij}(k) \cdot M_{ijh}(k)$ are the actual transfer flows. The perimeter control $u_{ij}(k)$ is defined as the ratio of vehicles that can actually transfer from region i to region j , which is relative to the transfer flow under the case of original fixed timing control (i.e., no control, NC) [24]. When the traffic is sufficient, the transfer flow is proportional to the green time ratio of boundary intersections. For simplicity, it is assumed that all the boundary intersections have the same signal period and timing plan [25]. Denote the initial fixed green time ratio of boundary intersections is ω_f , and that under perimeter control is ω_p . Then the physical meaning of perimeter control input is approximated as $u_{ij} = \omega_p / \omega_f$.

It is noticeable that there is a boundary capacity between every two neighboring regions. The boundary capacity $C_{ij}(n_j(k))$ is considered as below:

$$C_{ij}(n_j(k)) = \begin{cases} C_{ij}^{\max} & \text{if } 0 \leq n_j(k) \leq \alpha \cdot n_j^{\text{jam}} \\ \frac{C_{ij}^{\min} - C_{ij}^{\max}}{(1 - \alpha) \cdot n_j^{\text{jam}}} \cdot n_j(k) + \frac{C_{ij}^{\max} - \alpha \cdot C_{ij}^{\min}}{1 - \alpha} & \text{if } \alpha \cdot n_j^{\text{jam}} \leq n_j(k) \leq n_j^{\text{jam}} \end{cases} \quad (13)$$

where $C_{ij}^{\max}$ and $C_{ij}^{\min}$ represent the maximum and the minimum value of the boundary capacity, respectively, and $0 < \alpha < 1$. $C_{ij}^{\min}$ is equal to 0 in [13], [30] for simplification. In reality, however, there is still a very small amount of trip completion flow (i.e., $G_j(n_j^{\text{jam}})$, such as MFD in the downtown of Yokohama) under this circumstance. Therefore, in this work, $C_{ij}^{\min}$ is modified to a portion of $G_j(n_j^{\text{jam}})$, which is more realistic.

In consideration of the boundary capacity, one has $u_{ij}(k) \leq C_{ij}(n_j(k)) / M_{ij}(k)$. In addition, $u_{ij}(k)$ is also limited by the green time ratio. Denote the maximum and minimum green time ratios as $\omega_{\max}$ and $\omega_{\min}$, respectively. Thus, $u_{ij,\max}^g = \omega_{\max} / \omega_f$ and $u_{ij,\min}^g = \omega_{\min} / \omega_f$ are the theoretical approximation of maximum and minimum perimeter control ratio, respectively. Therefore, the range of perimeter control input is presented below:

$$u_{ij,\min}^g \leq u_{ij}(k) \leq \min\{u_{ij,\max}^g, C_{ij}(n_j(k)) / M_{ij}(k)\} \quad (14)$$

B. Route Choice (RC)

In the urban traffic system, RC is often used to describe the actual traffic situation [13]. Therefore, the traffic simulation system with route choice can be more closely related to the actual traffic flow status. The RC strategy is shown as follows.

For region $j \in N_i$, one has $b_{ijj}(k) = 1$ and $b_{ihj}(k) = 0$, which means that vehicles are suggested to enter region j directly. Hence, vehicles with destinations in neighboring regions no longer need to go the long way around.

For region $j \notin N_i$, a logit model [53] integrating with Dijkstra's algorithm [54] for K -shortest paths strategy (similar to [13], [30]) is utilized to calculate the route choice ratio

as below:

$$b_{ihj}^{\text{logit}}(k) = e^{\tau/t_{ihj}(k)} / \sum_{h \in V_{ij}} e^{\tau/t_{ihj}(k)} \quad (15)$$

$$b_{ihj}(k) = \beta \cdot b_{ihj}^{\text{init}} + (1 - \beta) \cdot b_{ihj}^{\text{logit}}(k), \quad j \notin N_i, h \in V_{ij} \quad (16)$$

where the definition of the above variables are listed below:

$b_{ihj}^{\text{logit}}(k)$	The logit ratio of vehicles from region i with destination to region j choosing to go through the next immediate region h at time step k ,
b_{ihj}^{init}	The initial ratio of vehicles from region i with destination to region j choosing to go through the next immediate region h ,
$t_{ihj}(k)$	The shortest average travel time for the route from region i to region j via the next immediate region h at time step k ,
τ	A positive scale parameter,
β	The rate of drivers who persist in their initial choices, $0 \leq \beta \leq 1$.

In fact, the routes those take a lot of time and need to go too far away (for example, region h locates at the upstream of region i in the direction to region j) are always not chosen by drivers. Therefore, the first K shortest routes are generally provided for drivers to choose from via (15).

III. METHODOLOGY FRAMEWORK

As shown in the above dynamics (1)–(16), the multi-region urban road network system is a complex interconnected strong-coupled MIMO nonlinear system. The control methods for MIMO nonlinear systems can be utilized for the system, but it will definitely lead to too complex control system if the model based control methods are used to design the controller, in a sequel, it is difficult to implement. In the next, the MIMO MRUTS is first decomposed into some multiple-input and single-output (MISO) subsystems, then the DED-MFAC for the complex interconnected system is proposed to the perimeter control for the MIMO nonlinear multi-region urban traffic system.

A. System Decomposition

In this section, the MRUTS is decomposed into N MISO subsystems through the urban road network is divided into N regions, shown in Fig. 1. The whole urban traffic system has N outputs, that is, the accumulation of each region, and there are two perimeter control inputs in each boundary between every two adjacent regions. For subsystem j , the controllable perimeter control inputs are u_{ij} , $i \in N_j$, and the output is n_j . Besides, u_{ji} , $i \in N_j$, are uncontrollable for subsystem j . It means that the inflow of region j is controllable for subsystem j , while the outflow of region j is uncontrollable. Instead, it is controlled by other subsystems.

For the j -th subsystem, when one perimeter control input u_{ij} is regarded as the controllable control input, the other perimeter control inputs related to subsystem j are considered as the interconnected influences.

The interconnected influence of subsystem h on subsystem j ($h \in N_j$) consists of two parts. On one hand, u_{jh} leads to the reduction of the output of subsystem j . On the other hand, u_{hj} results in the increase of it. The interaction of other subsystems acting on subsystem 1 is shown in Fig. 1 as an example. In this example, u_{21} is considered as the controllable perimeter control input of subsystem 1, and other related perimeter control inputs are the interconnected effects of other subsystems acting on subsystem 1. The same is true for other subsystems.

For all the regions $h \in N_j$, the subsystem h has the interconnected influence on subsystem j , denoted by z_{hj} . For $h \notin N_j$, one has $z_{hj} = 0$. The physical meaning of z_{hj} is the increment of vehicles contributed by region h to region j , i.e.

$$z_{hj}(k) = T \cdot (u_{hj}(k-1) \cdot M_{hj}(k-1) - u_{jh}(k-1) \cdot M_{jh}(k-1)) \quad (17)$$

$$z_{ij}(k) = -T \cdot u_{ji}(k-1) \cdot M_{ji}(k-1) \quad (18)$$

The reason why there is a minus sign “−” in (18) is that z_{ij} means the number of vehicles flowing out of region j to region i , leading to the reduction of n_j . Since u_{ij} is the controllable perimeter control input of subsystem j , the interconnected influence of subsystem i on subsystem j is only composed of u_{ji} , see (18) and Fig. 1. For subsystem j , the interactions (17)–(18) are uncontrollable but can be measured by traffic detectors.

B. Dynamic Linearization for Subsystems

After decomposition, each subsystem is still a nonlinear system. Fortunately, an equivalent description of the unknown nonlinear system can be provided by dynamic linearization data modeling technique. The dynamic linearization data modeling method is to describe the dynamic data relationship between the changes of all factors affecting the system output at the current time and the changes of the system output at the next time. It differs from the state space model, but it can replace the functions of the latter. The dynamic linearization data model has three forms [49], [55]: the compact form dynamic linearization (CFDL), the partial form dynamic linearization (PFDL), and the full form dynamic linearization (FFDL) data model. The time-varying dynamical relationship between the increments of the system output at the next time step and that of the control input at the current time step is considered in CFDL data modeling scheme. In PFDL data modeling method, the impacts on the increments of system output at the next time step imposed by the change of control input within a fixed length moving time window at the current time are all taken into consideration. In FFDL data modeling approach, all the effects on the system output increments at the next time step imposed by both the control input increments and the system output increments within input-related and output-related fixed length moving time windows at the current time step are fully taken into account. The above three forms

of data model are all applicable for the MRUTS. Without loss of generality, CFDL data model is utilized in this paper.

Assume that a complex interconnected MIMO nonlinear system consists of N subsystems, and the j -th subsystem is described as below:

$$\begin{aligned} y_j(k+1) &= f_j(y_j(k), \dots, y_j(k-n_{y_j}), u_{ij}(k), \dots, u_{ij}(k-n_{u_{ij}}), \\ &\quad z_{1j}(k), \dots, z_{1j}(k-n_{z_{1j}}), \dots, z_{hj}(k), \dots, \\ &\quad z_{hj}(k-n_{z_{hj}}), \dots, z_{Nj}(k), \dots, z_{Nj}(k-n_{z_{Nj}})), \\ i, j, h &= 1, 2, \dots, N, \quad i, h \in N_j \end{aligned} \quad (19)$$

where $u_{ij}(k) \in R$ and $y_j(k) \in R$ represent the controllable control input and the output of subsystem j , respectively. $z_{hj}(k) \in R$, $h \neq j$, is the measurable interactions of subsystem h acting on subsystem j . n_{y_j} , $n_{u_{ij}}$ and $n_{z_{hj}}$ are unknown integers, and $f_j(\dots)$ is an unknown nonlinear function, which describes the dynamics of subsystem j .

For the multi-region perimeter control problem (1)–(18), writing them with another compact form, it can be rewritten in following equation (20).

$$n_j(k+1) = f_j(n_j(k), u_{ij}(k), z_{hj}(k)), \quad i, h, j = 1, \dots, N \quad (20)$$

Comparing (19) and (20), we can see that they have the same descriptions, and n_{y_j} , $n_{u_{ij}}$, and $n_{z_{hj}}$ are all equal to 0 in the given multi-region perimeter control problem of this paper. In order to make the symbol notation simplification, and without loss the rigor, we still use $y_j(k)$ to represent $n_j(k)$.

Define the augmented control input vector of subsystem j with respect to u_{ij} as $\mathbf{u}_{ij}(k) = [u_{ij}(k), z_{1j}(k), \dots, z_{hj}(k), \dots, z_{Nj}(k)]^T$, $h \neq j$, $i, j, h = 1, \dots, N$, $z_{hj}(k) = 0$ if $h \notin N_j$. The j -th subsystem (19) can be rewritten as

$$y_j(k+1) = f_j(y_j(k), \dots, y_j(k-n_{y_j}), \mathbf{u}_{ij}(k), \dots, \mathbf{u}_{ij}(k-n_{u_{ij}})) \quad (21)$$

where $n_{u_{ij}} = \max\{n_{u_{ij}}, n_{z_{1j}}, \dots, n_{z_{hj}}, \dots, n_{z_{Nj}}\}$, $h \neq j$, $i, j, h = 1, \dots, N$.

The augmented control input vector of each subsystem j is composed of the controllable perimeter control u_{ij} and the measurable interactions from other subsystems. It is noted that the following two assumptions are satisfied for the considered complex interconnected MRUTS.

Assumption 1: The partial derivatives of $f_j(\dots)$ with respect to each component of the control vector $\mathbf{u}_{ij}(k)$ are continuous.

Assumption 2: Each subsystem (21) is generalized Lipschitz, i.e., $\|y_j(k_1+1) - y_j(k_2+1)\| \leq b \|u_{ij}(k_1) - u_{ij}(k_2)\|$ for all $k_1 \neq k_2$, $k_1, k_2 > 0$ and $\mathbf{u}_{ij}(k_1) \neq \mathbf{u}_{ij}(k_2)$, where $y_j(k_p+1) = f(y_j(k_p), \dots, y_j(k_p-n_{y_j}), \mathbf{u}_{ij}(k_p), \dots, \mathbf{u}_{ij}(k_p-n_{u_{ij}}))$, $p = 1, 2$, and b is a positive constant.

Remark 2: For the urban traffic system, assumption 1 is easy to be verified from the dynamics of the MRUTS (1)–(20), and meanwhile it is also a typical assumption for controller design for nonlinear systems. Assumption 2 is a physical constraint by the inherent nature of urban traffic system, i.e., finite change of vehicle flow does not lead to infinite change of the number of vehicles in a region. Meanwhile, it is also a physical constraint of the real system from the energy point of view.

Theorem 1 [49]: Consider the nonlinear subsystem j , satisfying the assumptions 1 and 2 and $\|\Delta \mathbf{u}_{ij}(k)\| \neq 0$, there must exist a pseudo gradient (PG) denoted by $\phi_{ij}(k)$, such that the subsystem (21) can be transformed into the following equivalent CFDL data model:

$$\Delta y_j(k+1) = \phi_{ij}^T(k) \Delta \mathbf{u}_{ij}(k) \quad (22)$$

where $\phi_{ij}(k) = [\phi_{ij}(k), \phi_{1j}(k), \dots, \phi_{hj}(k), \dots, \phi_{Nj}(k)]^T$, $h \neq j$, $i, j, h = 1, \dots, N$, is the PG of the subsystem j with respect to u_{ij} , $\Delta y_j(k+1) = y_j(k+1) - y_j(k)$, and $\Delta \mathbf{u}_{ij}(k) = \mathbf{u}_{ij}(k) - \mathbf{u}_{ij}(k-1)$.

Remark 3: It can be seen from (22) that the change of the j -th subsystem's output $\Delta y_j(k+1)$ is not only related to the change of the controllable control input of subsystem j (i.e., $u_{ij}(k)$), but also influenced by the change of the interactions from other subsystems, which are depicted by $\phi_{ij}(k) \Delta \mathbf{u}_{ij}(k)$ and $\sum_{h=1, h \neq j}^N \phi_{hj}(k) \cdot \Delta z_{hj}(k)$, respectively. The latter item fully illustrates the impact of other subsystems acting on subsystem j . In the MRUTS, the physical meaning of the above discussion is that the change of the accumulations in region j is not only related to the change of the controllable perimeter control ratio $u_{ij}(k)$, which controls the inflow from the corresponding region i , but also associated with the change of the inflow from and the outflow to other regions, which are measurable.

C. DED-MFAC Perimeter Controller Design

For the convenience of reading, a brief description for DED-MFAC is introduced here. After CFDL for the MISO nonlinear subsystem j , the perimeter control of DED-MFAC scheme is designed as follows.

The objective function for estimating PG is as follows:

$$J_j(\phi_{ij}(k)) = \left| y_j(k) - y_j(k-1) - \phi_{ij}^T(k) \Delta \mathbf{u}_{ij}(k-1) \right|^2 + \mu \left\| \phi_{ij}(k) - \hat{\phi}_{ij}(k-1) \right\|^2 \quad (23)$$

where $\mu > 0$ is a weighting factor to penalize excessive change of the estimation of the PG. PG updating algorithm is then designed by minimizing (23) via some modifications with consideration of practical applications:

$$\hat{\phi}_{ij}(k) = \hat{\phi}_{ij}(1), \text{ if } \left| \hat{\phi}_{ij}(k) \right| < \varepsilon \text{ or } \text{sign}(\hat{\phi}_{ij}(k)) \neq \text{sign}(\hat{\phi}_{ij}(1)) \quad (25)$$

$$\hat{\phi}_{hj}(k) = \hat{\phi}_{hj}(1), \text{ if } \left| \hat{\phi}_{hj}(k) \right| < \varepsilon \text{ or } \text{sign}(\hat{\phi}_{hj}(k)) \neq \text{sign}(\hat{\phi}_{hj}(1)) \quad (26)$$

where $\hat{\phi}_{ij}(1)$ and $\hat{\phi}_{hj}(1)$ are the initial values of $\hat{\phi}_{ij}(k)$ and $\hat{\phi}_{hj}(k)$, $h \neq j$, $i, j, h = 1, \dots, N$, respectively, and $\hat{\phi}_{ij}(k)$ is the estimation of $\phi_{ij}(k)$.

The objective function for perimeter control input design is as follows:

$$J_j(u_{ij}(k)) = \left| y_j^*(k+1) - y_j(k+1) \right|^2 + \zeta \lambda_j \left| u_{ij}(k) - u_{ij}(k-1) \right|^2 \quad (27)$$

where $y_j^*(k+1)$ (i.e., $n_j^*(k+1)$) is the desired output signal of subsystem j , which represents the expected number of vehicles in region j . $\lambda_j > 0$ is a weighting constant to penalize excessive changes of the control input signal, i.e., the perimeter control ratios should not change too fast. This is the practical consideration for traffic control. If the green light time of the two consecutive cycles changes too fast, it will result in some risk of safety. ζ is a factor that makes the order of magnitude of the two items in (27) are in the same level.

Substituting (22) into (27), then differentiating it with respect to $u_{ij}(k)$, and letting it zero, gives the perimeter control ratio as below:

where $\eta \in (0, 2]$ and $\rho \in (0, 1]$ are step-size constants, and ε is a positive constant. (29) is extra added in the control input algorithm for practical consideration of traffic control application, which has the same functions as (14) to deal with the actual traffic situation.

The DED-MFAC scheme consists of (24)–(26) and (28)–(29), shown at the top of the next page, where the reset mechanism (25)–(26) are designed to make the parameter estimation algorithm (24), shown at the top of the next page, have stronger ability for tracking time-varying parameters of the traffic system. For other discussions on the DED-MFAC scheme, please refer to [49] for details.

The outstanding features of the proposed DED-MFAC scheme are as follows:

1) Since that the MIMO nonlinear urban traffic system is decomposed to N subsystems, and each subsystem is a MISO nonlinear system with only one controllable input and $N-1$ uncontrollable inputs, in which the latter is regarded as the interconnected actions from other subsystems. Fortunately, it has no difficulty for traffic practice since these influences are easily measured using geomagnetic coil or video detector. The interactions among the subsystems are compensated via the last item in the numerator of the second item on the right side of the equal sign in (28), so that the internal coupling of the MIMO nonlinear systems can be addressed by the DED-MFAC approach. By establishing CFDL data model with measurable interactions and designing MFAC scheme for each subsystem, the decentralized estimation and decentralized control for the whole MRUTS is realized.

2) The proposed DED-MFAC perimeter control strategy is independent of the mathematical model of the urban traffic system. Instead, only the input and output data of the urban traffic system (i.e., the perimeter control ratios at the last time and the accumulations at the current time) are needed in the process of perimeter controller design. Besides, there is no unmodeled traffic system dynamics in the proposed DED-MFAC scheme, and the structure and the orders of the mathematical model of the urban traffic system are both no longer required to be known. Furthermore, the DED-MFAC method does not need any training process or satisfy the persistence excitation conditions, which are usually necessary for other adaptive control methods [26], [27] for urban traffic control.

3) There are different limitations in other typical linearization methods for perimeter control. For instance, the high-order terms are ignored for the Taylor's linearization [25],

$$\hat{\phi}_{ij}(k) = \hat{\phi}_{ij}(k-1) + \frac{\eta \Delta \mathbf{u}_{ij}(k-1)(y_j(k) - y_j(k-1) - \hat{\phi}_{ij}^T(k-1) \Delta \mathbf{u}_{ij}(k-1))}{\mu + \|\Delta \mathbf{u}_{ij}(k-1)\|^2} \quad (24)$$

$$u_{ij}(k) = u_{ij}(k-1) + \frac{\rho \hat{\phi}_{ij}(k)(y_j^*(k+1) - y_j(k) - \sum_{h=1, h \neq j}^N \hat{\phi}_{hj}(k) \Delta z_{hj}(k))}{\zeta \lambda_j + \hat{\phi}_{ij}^2(k)} \quad (28)$$

$$u_{ij}(k) = \begin{cases} u_{ij,\min}^g, & \text{if } u_{ij}(k) < u_{ij,\min}^g \\ u_{ij}(k), & \text{if } u_{ij,\min}^g \leq u_{ij}(k) \leq \min\{u_{ij,\max}^g, C_{ij}((n_j(k))/M_{ij}(k))\} \\ \min\{u_{ij,\max}^g, C_{ij}((n_j(k))/M_{ij}(k))\}, & \\ u_{ij}(k), & \text{if } u_{ij}(k) > \min\{u_{ij,\max}^g, C_{ij}((n_j(k))/M_{ij}(k))\} \end{cases} \quad (29)$$

more model information of the urban traffic system is required for the piecewise linearization [22], such as the switching and residence time of the piecewise linearized dynamics of the urban traffic system. Further, they are both difficult to handle the parameter uncertainties or disturbances in MFD, demand, and accumulation measurement, etc. Different from the above existing linearization methods, an equivalent description of the original MIMO urban traffic system is provided by CFDL data model, in which only the input and output data of the MRUTS is needed, and the system dynamics is no longer required.

4) The existence of the PG is guaranteed by rigorous mathematical analysis [49]. Meanwhile, the CFDL data model is very simple since the PG is easily estimated by using the urban traffic system measurement data, without heavy computational burden. Finally, since the PG is a differential signal in some sense, and is a slowly time-varying parameter for the practical traffic system, thus it is insensitive to the time-varying factors of the urban traffic system, such as accumulations, demands, etc. These features will make the DED-MFAC method have strong robustness.

However, if the MIMO system cannot be decomposed into subsystems with measurable interactions, or the system is too complex that it cannot be separated into N subsystems, then the proposed control method will no longer work well. This is the main limitation of the proposed method.

IV. SIMULATION RESULTS

Two cases are simulated to test the performance of the proposed DED-MFAC strategy for the multi-region perimeter control problem. In case I, the measurement noises in demands, accumulations, and MFDs are integrated to test the effectiveness of the proposed perimeter control strategy with uncertainties in the traffic model. In addition, no control (NC) [24] and model predictive control (MPC) [22], [30] are also applied to the MRUTS under the same condition with the same route choice strategy to verify the superiority of DED-MFAC perimeter control strategy. In case II, the measurement noises are also added to the measurable interactions under DED-MFAC strategy to further study the robustness of it.

In the simulation, the following MFD-based urban traffic model is only used to produce the traffic data, rather than in the process of perimeter controller design.

A. The Setting of Parameters and Traffic Demands

The urban traffic network is partitioned into five regions (i.e., $N = 5$) in the simulation. Region 1 is at the center of the city, while region 2–5 are on the periphery. In order to make the simulation simple and time-saving, it is assumed that the structure of MFD in each region is the same, while the parameters are different. Region 2–5 have the same MFD as the one in the downtown of Yokohama [11], [22], while the MFD of region 1 is 1.2 times as the ones in other regions. The trip completion flow $G_i(n_i(k))$ is approximately formulated as a third-order polynomial function of $n_i(k)$, i.e.

$$G_i(n_i(k)) = \begin{cases} (1.7852 \cdot 10^{-7} \cdot n_i(k)^3 - 3.5022 \cdot 10^{-3} \cdot n_i(k)^2 \\ \quad + 18.1094 \cdot n_i(k))/3600, & i = 1. \\ (1.4877 \cdot 10^{-7} \cdot n_i(k)^3 - 2.9185 \cdot 10^{-3} \cdot n_i(k)^2 \\ \quad + 15.0912 \cdot n_i(k))/3600, & i = 2, 3, 4, 5. \end{cases} \quad (30)$$

where $n_i^{cr} = 3400$ (veh), $G_i(n_i^{cr}) = 6.3$ (veh/s), $n_i^{jam} = 10000$ (veh), and $G_i(n_i^{jam}) = 0.43$ (veh/s), $i = 2, 3, 4, 5$, while for region 1, one has $G_1(n_1^{cr}) = 7.56$ (veh/s) and $G_1(n_1^{jam}) = 0.52$ (veh/s), and the other MFD parameters are the same as the ones in the MFD of region 2–5. The parameters of boundary capacity are $\alpha = 0.64$, $C_{ij}^{\max} = 3.2$ (veh/s) [30], and $C_{ij}^{\min} = 0.15$ (veh/s) for $j = 2, 3, 4, 5$ and $i \in N_j$, while for region 1, the corresponding parameters are $C_{i1}^{\max} = 3.84$ (veh/s) and $C_{i1}^{\min} = 0.18$ (veh/s), $i \neq 1$, respectively.

The initial number of vehicles are: $n_1(0) = 5000$, $n_2(0) = 4000$, $n_3(0) = 4000$, $n_4(0) = 4000$, and $n_5(0) = 4000$, respectively. All the regions are initially congested according to the definition of MFD, since region i is congested when the accumulation is greater than n_i^{cr} . The initial conditions are set up to simulate the actual situation of morning peak. The initial

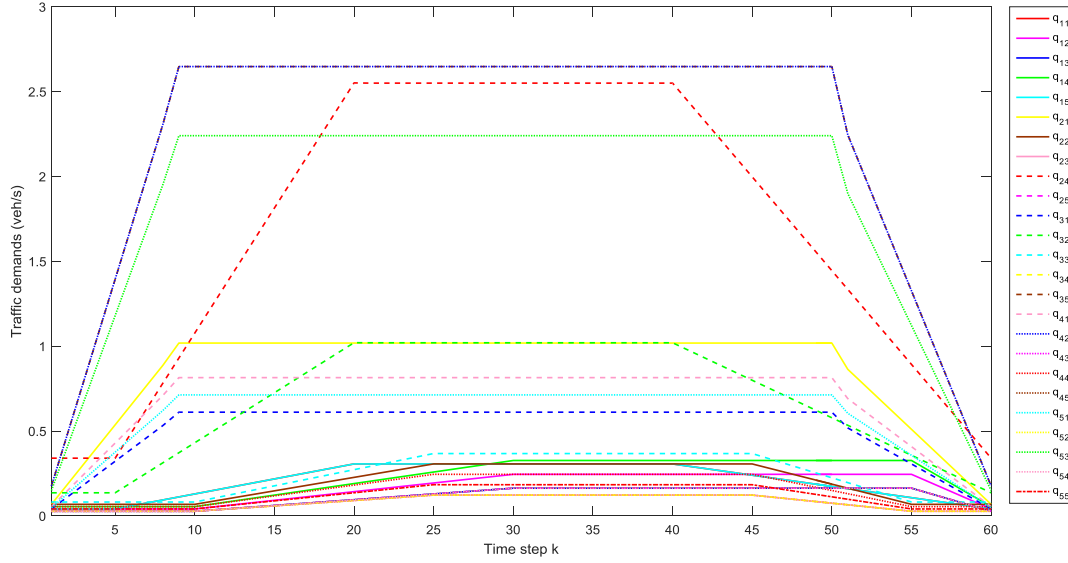


Fig. 2. Traffic demands for each pair of regions during the whole simulation period.

green time ratio is $\omega_f = 0.5$, while the maximum and the minimum green time ratios are $\omega_{\max} = 0.9$ and $\omega_{\min} = 0.1$, respectively. According to the definition of perimeter control ratio, one has $u_{ij,\max}^g = 1.8$ and $u_{ij,\min}^g = 0.2$.

The initial route choice ratios are $b_{ij}^{\text{init}} = 0.8$ and $b_{ihj}^{\text{init}} = 0.1$ respectively, $i, j, h = 2, 3, 4, 5, h \in V_{ij}$. Other parameters of route choice are $\tau = 50000$, $\beta = 0.3$, and $K = 3$, respectively.

The measurement noises in the accumulations, demands, MFDs, and measurable interactions obey normal and uniform distribution, respectively, which are presented as follows:

$$\tilde{n}_i(k) = n_i(k) \cdot (1 + \mathbb{N}(0, \sigma_{n_i}^2)) \quad (31)$$

$$\tilde{q}_{ij}(k) = q_{ij}(k) \cdot (1 + \mathbb{N}(0, \sigma_{q_{ij}}^2)) \quad (32)$$

$$\tilde{G}_i(n_i(k)) = G_i(n_i(k)) \cdot (1 + \mathbb{U}(-\alpha_{G_i}, \alpha_{G_i})) \quad (33)$$

$$\tilde{z}_{ij}(k) = z_{ij}(k) \cdot (1 + \mathbb{N}(0, \sigma_{z_{ij}}^2)) \quad (34)$$

where the error parameters are chosen as $\sigma_{n_i}^2 = 0.1$, $\sigma_{q_{ij}}^2 = 0.1$, and $\alpha_{G_i} = 0.35$ in both cases, respectively. In the second case, the additional noise parameter of the measurable interactions is selected as $\sigma_{z_{ij}}^2 = 0.1$.

The simulation duration is set as 4 hours including the morning peak hour, and the signal cycle length of the boundary intersections is set as 120s, while the control cycle length is $T = 240$ s, which is the common multiple of the signal cycle length for control convenience. Then the 4 hours of simulation duration is divided into 60 time steps (i.e., $k^{\text{end}} = 60$). The time-varying traffic demands are shown in Fig. 2 to simulate the real traffic situation in the morning peak hour. The simulation is based on MATLAB, and the CPU configuration is Intel(R) Core(TM) i7-6700HQ, 2.60GHz.

The following four evaluation indicators, i.e., the total network throughput (TNT) for the entire urban road network, the total time spent (TTS) and the average travel time (ATT) for all vehicles, and the total CPU time (TCT), are utilized to evaluate the effectiveness of the perimeter control methods. The first three indicators are defined as

follows: $TNT = T \cdot \sum_{k=1}^{k^{\text{end}}} \sum_{i=1}^N M_{ii}(k)$, $TTS = T \cdot \sum_{k=1}^{k^{\text{end}}} \sum_{i=1}^N n_i(k)$, and $ATT = \frac{TTS}{TNT}$, respectively, and TCT means the total time for calculating the perimeter control inputs during the simulation. Under the same initial condition and traffic demands, the more TTP, and the less TTS, ATT, and TCT, the better performance the perimeter control strategy has.

B. Perimeter Control Methods

1) *DED-MFAC*: In the proposed DED-MFAC strategy, the estimated value of the PGs and the perimeter control ratios are obtained via (24)–(26) and (28)–(29), respectively. The urban traffic model (1)–(16) are utilized to generate the urban traffic data, instead of controller design. The desired output of each subsystem i is chosen as $0.98 \cdot n_i^{\text{cr}}$ in order to make the trip completion flow near the maximum value without congestion. Other parameters are: $\zeta = 1000$, $\lambda_i = 10$, $\eta = 1$, $\rho = 1$, and $\mu = 1$, respectively.

2) *No Control (NC)*: In the case of NC, the perimeter control ratio u_{ij} is constant to 1 [24]. NC is served as a benchmark for comparing the perimeter control effects.

3) *Model Predictive Control (MPC)*: MPC [22], [30] is a frequently used perimeter control method benefiting from its excellent control performance in the case of models are precisely known. In this paper, MPC serves as a model based traffic control method for the performance comparison. The urban traffic model proposed in Section II is utilized to design the MPC perimeter controller. The objective functions and the constraints of the MPC strategy are the same as the ones in [22]. The predictive horizon and the control horizon are chosen as $N_p = 3$ and $N_c = 2$, respectively.

C. Simulation Results

1) *Case I*: The simulation results under case I are presented in Fig. 3–5 and the first three lines of Table I respectively.

TABLE I
EVALUATION RESULTS OF DIFFERENT STRATEGIES UNDER DIFFERENT CASES

Case	Strategy	Total Network Throughput TNT (veh)	Total Time Spent TTS (s)	Average Travel Time ATT (s)	Total CPU Time TCT (s)
I	NC	6.4353×10^4	3.8798×10^8	6.0288×10^3	0.25
I	MPC	1.9208×10^5	2.1497×10^8	1.1192×10^3	174.82
I	DED-MFAC	2.1923×10^5	1.6720×10^8	762.70	0.46
II	DED-MFAC	2.1938×10^5	1.7876×10^8	814.31	0.48

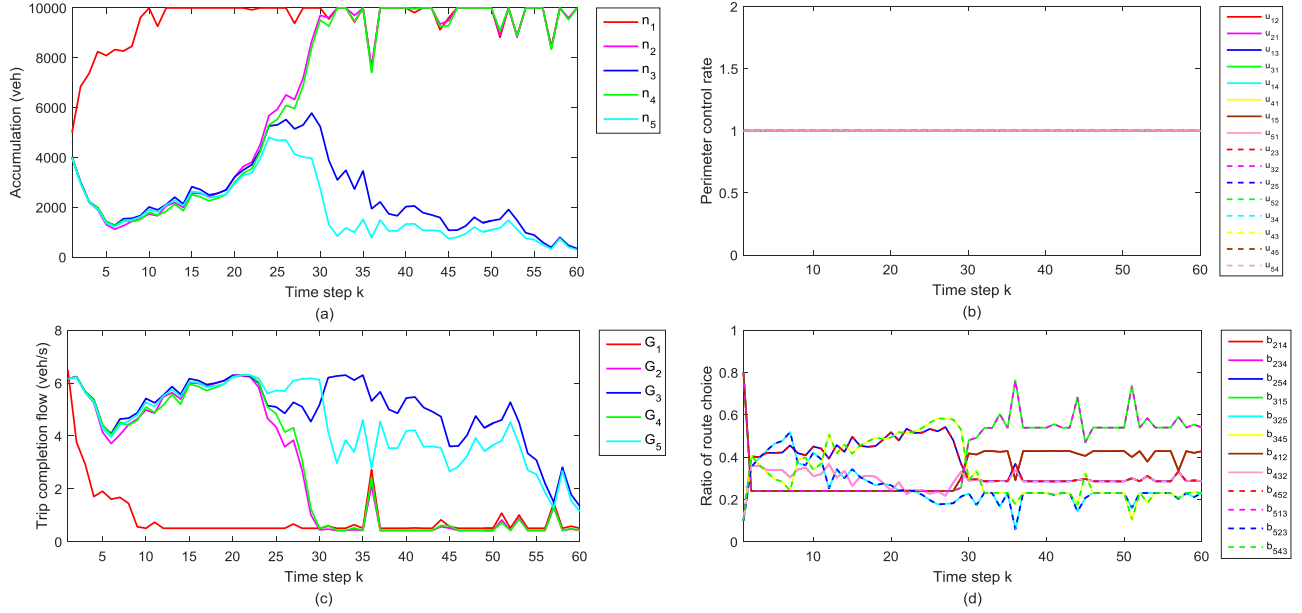


Fig. 3. Result of NC strategy under case I: (a) evolution of accumulation in each region; (b) perimeter control input of NC; (c) trip completion flow for each region; (d) ratio of route choice.

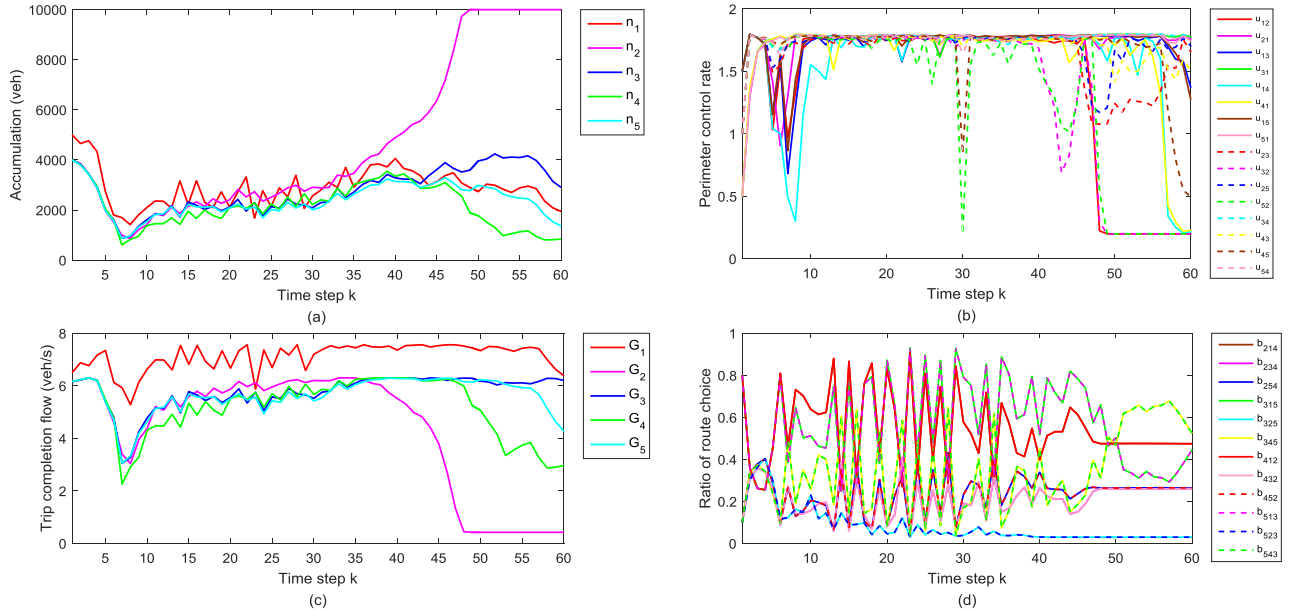


Fig. 4. Result of MPC strategy under case I: (a) evolution of accumulation in each region; (b) perimeter control input of MPC; (c) trip completion flow for each region; (d) ratio of route choice.

The evolution of accumulations in the five regions under different perimeter control strategies are depicted in Fig. 3 (a) –Fig. 5 (a), respectively. It can be seen from Fig. 3 (a) that there was a long period of jam phenomenon

in 3 of the 5 regions under NC strategy. MPC worked better than NC, since that there was only one region congested at the end of the morning peak, while the other 4 regions were well operated since the accumulations of them were changing

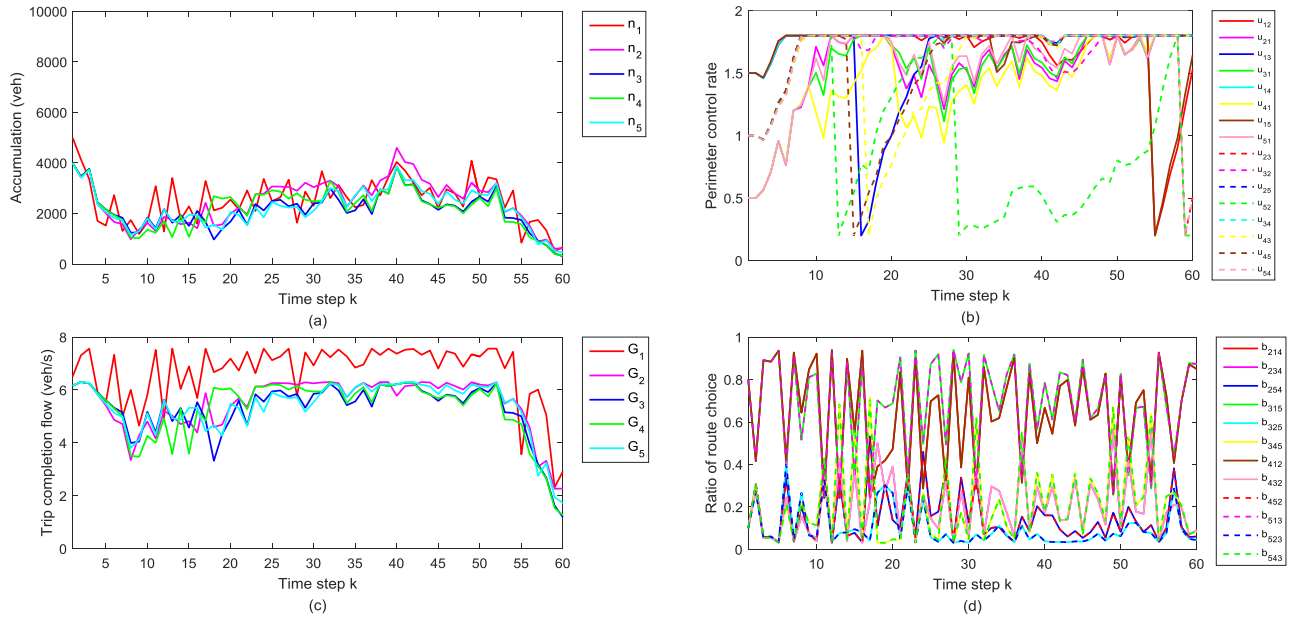


Fig. 5. Result of DED-MFAC strategy under case I: (a) evolution of accumulation in each region; (b) perimeter control input of DED-MFAC; (c) trip completion flow for each region; (d) ratio of route choice.

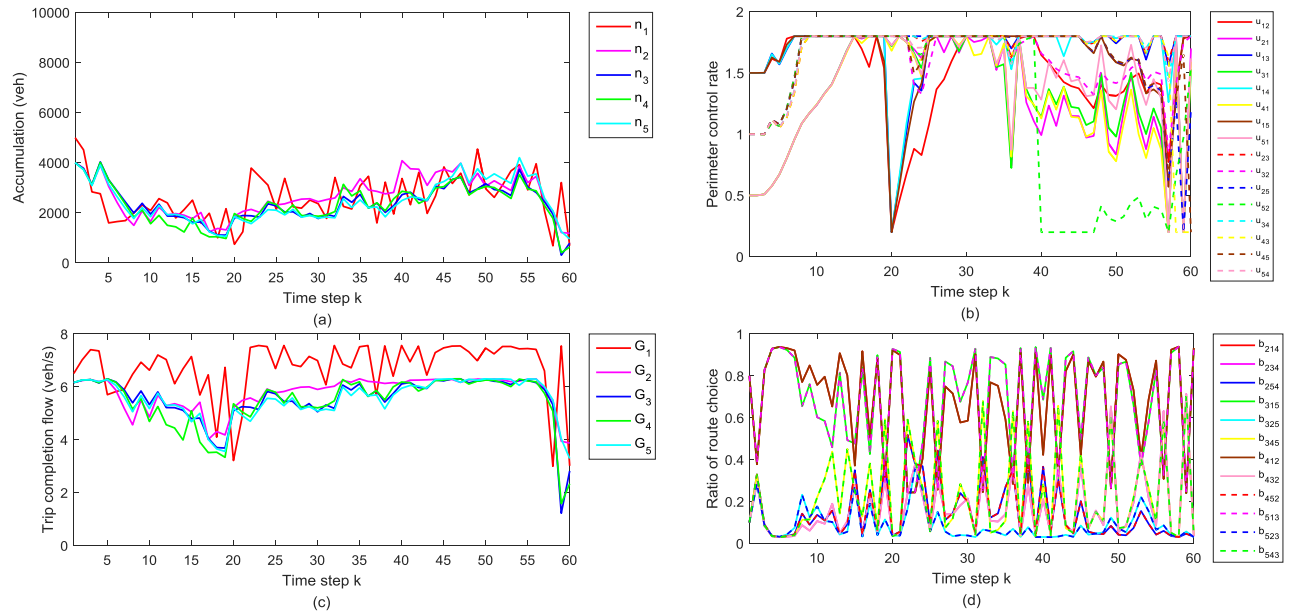


Fig. 6. Result of DED-MFAC strategy under case II: (a) evolution of accumulation in each region; (b) perimeter control input of DED-MFAC; (c) trip completion flow for each region; (d) ratio of route choice.

around the critical values, see Fig. 4 (a). On the contrary, it can be seen from Fig. 5 (a) that all the five regions were no longer congested all the way under DED-MFAC approach, and the accumulations in each region kept near the desired values during the peak hour. The time-varying perimeter control ratios and route choice ratios of the above strategies are displayed in Fig. 3 (b)–Fig. 5 (b) and Fig. 3 (d)–Fig. 5 (d), respectively.

The trip completion flows for the regions under each perimeter control strategy are shown in Fig. 3 (c)–Fig. 5 (c), respectively. As depicted in the figures, the trip completion flows maintained at a relatively high level under DED-MFAC

strategy during the morning peak since the accumulations in each region were near the expected values, while fewer vehicles in the congested regions could finish their trips under the other strategies because of the fact that the trip completion flow will decrease when the accumulation is greater than the critical value.

The first three lines of Table I indicate that DED-MFAC is superior to other strategies for the multi-region perimeter control problem, as the TNT is greater than others, while the TTS, ATT, and TCT are less than other perimeter control strategies.

2) *Case II*: The simulation results of case II are presented in Fig. 6 and the last line of Table I, respectively. It can be seen from Fig. 6 (a) that although the noises were added into the measurable interactions, the number of vehicles in each region were still operated near the expected values, and the trip completion flows of the five regions still remained near their maximum values, see Fig. 6 (c). The perimeter control inputs and the route choice ratios under this case are depicted in Fig. 6 (b) and (d), respectively. The last line of Table I reveals that the proposed DED-MFAC perimeter control method is also superior to other strategies in spite of the uncertainties in the measurable interactions. Meanwhile, it is noteworthy that the total CPU time 0.48s of DED-MFAC for calculating the control inputs is much smaller than the signal cycle 120s, which implies that it can be implemented in real time.

V. CONCLUSION

A novel data driven approach called DED-MFAC is first applied to deal with the problem of perimeter control for multi-region urban traffic network systems in this paper. The DED-MFAC scheme includes the following three parts: Firstly, the MIMO nonlinear multi-region urban traffic system is decomposed into N MISO subsystems. Secondly, CFDL data model for each subsystem is established. Finally, the PG estimation and perimeter controller design for each subsystem are proposed respectively. A remarkable advantage of the proposed method is that only the input and output data of the urban traffic system is needed to design the perimeter controller, and the urban traffic model is no longer required. The superiority and effectiveness of the proposed method are verified by comparison with MPC method via simulation. The results revealed that the performance of DED-MFAC strategy is superior to other methods with measurement noises in demands, accumulations, and MFDs, even when measurement noises appear in the interactions.

In the future work, we will focus on how to determine the desired number of vehicles in each region or in each link or even for an intersection via urban traffic big data directly. Meanwhile, we will try to make it implemented in some regions in Beijing.

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